SAT Math Final Revision Set 1

76 questions · 2 hours · Answers and solutions

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (13 items)

1. A-316 · Hard · SPR · **No calculator needed**

The function f is linear, $f(3) = k$ and $f(k) = 3$, where k is a constant and $k \neq 3$. If $f(10) = -1$, what is the value of k?

Answer: 6

The two points are $(3, k)$ and $(k, 3)$. Slope $= \dfrac{3-k}{k-3} = -1$ for every $k \neq 3$. So $f(x) = -x + b$. From $f(10) = -1$: $b = 9$, so $f(x) = -x + 9$ and $k = f(3) = 6$. Check: $f(6) = 3$ ✓.

Trap: Students think two unknowns (k and the slope) cannot be found, or assume the slope is $+1$ because the inputs and outputs are swapped.

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2. A-318 · Hard · SPR · **No calculator needed**

The function f is linear, $f(2) + f(8) = 30$, and $f(11) = 33$. What is the value of $f(0)$?

Answer: 0

For a linear function the average of two outputs is the output at the average input: $f(5) = \dfrac{30}{2} = 15$. Slope $= \dfrac{33-15}{11-5} = 3$, so $f(0) = 15 - 5 \cdot 3 = 0$.

Trap: Students think a sum of two outputs gives no usable point and stop, or use $(2, 30)$ as a point.

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3. A-321 · Hard · Multiple choice · **No calculator needed**

The system $2y - x > 4$ and $y < 5$ is graphed in the xy-plane. Which inequality describes all possible x-coordinates of points in the solution region?

A) $x > 6$

B) $x < 14$

C) $x < 6$

D) $x < -4$

Answer: (C) $x < 6$

From the first inequality, $x < 2y - 4$. Since $y < 5$, $2y - 4 < 6$, so every solution has $x < 6$. Every $x < 6$ is reached: for $x < 6$ choose y with $\dfrac{x+4}{2} < y < 5$.

Trap: Students solve each inequality on its own (for example $x < -4$ from $y = 0$) instead of combining them through y.

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4. A-323 · Hard · SPR · **No calculator needed**

If $-3 \leq x \leq 2$ and $-5 \leq y \leq 1$, what is the greatest possible value of xy?

Answer: 15

Check the corner products: $(-3)(-5) = 15$, $(-3)(1) = -3$, $2(-5) = -10$, $2(1) = 2$. The greatest is 15, from two negatives.

Trap: Students multiply the two upper bounds and answer 2.

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5. A-328 · Hard · SPR · **No calculator needed**

In the system $y = 2x + k$ and $y = -x + 10$, k is a constant. For how many integer values of k is the solution (x, y) in Quadrant I (that is, $x > 0$ and $y > 0$)?

Answer: 29

$2x + k = -x + 10 \Rightarrow x = \dfrac{10 - k}{3}$ and $y = \dfrac{20 + k}{3}$. $x > 0 \Rightarrow k < 10$; $y > 0 \Rightarrow k > -20$. Integers from -19 to 9: 29 values.

Trap: Students include an endpoint ($k = 10$ or $k = -20$) or count $9 - (-19) = 28$.

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6. A-332 · Hard · Multiple choice · **No calculator needed**

Consider the system $3x - 4y = 10$ and $-6x + 8y = -20$. Which point lies on the graphs of both equations for every real number t?

A) $\left(t, \dfrac{3t - 10}{4}\right)$

B) $\left(t, \dfrac{10 - 3t}{4}\right)$

C) $\left(t, \dfrac{3t + 10}{4}\right)$

D) $\left(\dfrac{4t - 10}{3}, t\right)$

Answer: (A) $\left(t, \dfrac{3t - 10}{4}\right)$

The second equation is -2 times the first, so the graphs are the same line and every point of $3x - 4y = 10$ works. Let $x = t$: $4y = 3t - 10$, so the point is $\left(t, \dfrac{3t - 10}{4}\right)$. (With $y = t$, $x = \dfrac{4t + 10}{3}$, not $\dfrac{4t - 10}{3}$.)

Trap: Students make a sign error isolating y, or give up because the system has "no single solution".

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7. A-335 · Hard · SPR · **No calculator needed**

An oven's temperature is $70°$F when it is switched on at $t = 0$. For $3 \le t \le 12$, where t is the time in minutes, the temperature is modeled by $T = a + bt$, where a and b are constants. The model gives $T = 200$ at $t = 3$ and $T = 380$ at $t = 7$. What is the value of a?

Answer: 65

$b = \dfrac{380 - 200}{7 - 3} = 45$. $a = 200 - 45(3) = 65$. The model is valid only for $t \ge 3$, so its intercept 65 need not equal the real starting temperature 70.

Trap: Students assume the intercept must be the given starting value 70.

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8. A-338 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the graph of $3x - ky = 12$, where k is a nonzero constant, has an x-intercept equal to its y-intercept. What is the value of k?

A) -4

B) 3

C) 4

D) -3

Answer: (D) -3

x-intercept: $3x = 12$, $x = 4$. y-intercept: $-ky = 12$, $y = -\dfrac{12}{k}$. Set $-\dfrac{12}{k} = 4$: $k = -3$.

Trap: Students drop the negative sign in $-ky$ and get $k = 3$.

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9. A-340 · Hard · Multiple choice · **No calculator needed**

A cafe sells juice for 4 dollars each and sandwiches for 7 dollars each. A customer spends exactly 112 dollars and buys at least one juice and at least one sandwich. How many different combinations of juices and sandwiches are possible?

A) 5

B) 4

C) 3

D) 28

Answer: (C) 3

$4x + 7y = 112$ with $x, y \geq 1$ integers. $4x = 112 - 7y$ needs y to be a multiple of 4: $y = 4, x = 21$; $y = 8, x = 14$; $y = 12, x = 7$ ($y = 16$ gives $x = 0$, not allowed). 3 combinations.

Trap: Students include $(28, 0)$ or $(0, 16)$ (answers 4 or 5) or divide 112 by 4.

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10. A-319 · Medium · Multiple choice · **No calculator needed**

In the xy-plane, what is the y-intercept of the graph of $x = 4y - 12$?

A) $(0, -12)$

B) $(0, -3)$

C) $(-12, 0)$

D) $(0, 3)$

Answer: (D) $(0, 3)$

Set $x = 0$: $4y = 12$, so $y = 3$. The y-intercept is $(0, 3)$. (Solving for y: $y = \frac{1}{4}x + 3$, so the slope is $\frac{1}{4}$, not 4.)

Trap: Students read the equation as if it were solved for y and take -12 as the y-intercept or 4 as the slope.

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11. A-325 · Medium · Multiple choice · **No calculator needed**

A line in the xy-plane has slope 3 and passes through the points $(2, k)$ and $(k, 14)$. What is the value of k?

A) 8

B) 4

C) 11

D) 5

Answer: (D) 5

$\frac{14 - k}{k - 2} = 3 \Rightarrow 14 - k = 3k - 6 \Rightarrow k = 5$. Check: points $(2, 5)$ and $(5, 14)$, slope $\frac{9}{3} = 3$ ✓.

Trap: Students write the slope as $\frac{k - 2}{14 - k}$ (run over rise) and get a wrong value such as 11.

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12. A-329 · Medium · Multiple choice · **No calculator needed**

In the inequality $kx + 4 < 10$, k is a constant and $k < 0$. Which describes the solution set?

A) $x < \frac{6}{k}$

B) $x > \frac{6}{k}$

C) $x > -\frac{6}{k}$

D) $x < -\frac{6}{k}$

Answer: (B) $x > \frac{6}{k}$

$kx < 6$. Dividing by the negative k reverses the inequality: $x > \frac{6}{k}$. Test $k = -2$: $-2x < 6 \Rightarrow x > -3 = \frac{6}{-2}$ ✓.

Trap: Students divide by k without reversing the sign because the letter k looks positive.

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13. **A-334** · Medium · Multiple choice · **No calculator needed**

How many solutions does the system $2x + y = 11$, $x - y = 1$, $x + 3y = -13$ have?

A) Exactly one

B) Exactly three

C) Zero

D) Exactly two

Answer: (C) Zero

The first two equations give $3x = 12$, so $(4, 3)$. Test it in the third: $4 + 3(3) = 13 \neq -13$. No point lies on all three lines: 0 solutions.

Trap: Students count the pairwise intersections (3) or stop after the first two equations (1).

Advanced Math (15 items)

14. **AM-309** · Hard · Multiple choice · **Desmos**

How many solutions (x, y) does the system $x^2 + y^2 = 25$ and $y = x^2 - 5$ have?

A) 3

B) 4

C) 1

D) 2

Answer: (A) 3

Substitute $x^2 = y + 5$: $y + 5 + y^2 = 25 \Rightarrow y^2 + y - 20 = 0 \Rightarrow y = 4$ or $y = -5$. $y = 4$: $x = \pm 3$ (two points). $y = -5$: $x^2 = 0$, $x = 0$ (one point). Total 3. Desmos: graph both and count the intersection points.

Trap: Students give every y-value two x-values (4) or count y-values only (2).

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15. **AM-317** · Hard · Multiple choice · **Desmos**

In the xy-plane, the graphs of $y = |x - 2|$ and $y = mx + 1$, where m is a constant, intersect at exactly two points. Which of the following could be the value of m?

A) -2

B) $-\dfrac{3}{4}$

C) 2

D) $\dfrac{1}{2}$

Answer: (D) $\dfrac{1}{2}$

Right branch ($x \geq 2$): $x = \dfrac{3}{1 - m} \geq 2$ needs $-\frac{1}{2} \leq m < 1$. Left branch ($x < 2$): $x = \dfrac{1}{1 + m} < 2$ needs $m > -\frac{1}{2}$ (or $m < -1$). Both branches meet the line when $-\frac{1}{2} < m < 1$, so $m = \frac{1}{2}$. Desmos: add a slider for m and count intersections.

Trap: Students think any line through $(0, 1)$ with a slope between -1 and 1 works, or check only one branch.

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16. **AM-303** · Hard · Multiple choice · **No calculator needed**

In the equation $(k - 2)x^2 + 4x + 1 = 0$, k is a constant. The equation has exactly one real solution for exactly two values of k. What is the sum of these two values of k?

A) 8

B) 4

C) 2

D) 6

Answer: (A) 8

Case 1: $k = 2$ makes the equation linear, $4x + 1 = 0$, one solution. Case 2: $k \neq 2$ and $\Delta = 16 - 4(k-2) = 0$, so $k = 6$. Sum $= 2 + 6 = 8$.

Trap: Students use only the discriminant and miss that a zero leading coefficient makes the equation linear.

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17. **AM-305** · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x) = \dfrac{x^2 - 4x - 12}{x^2 - 36}$. What is the sum of all values of x for which $h(x) = 0$?

A) 4

B) -8

C) 6

D) -2

Answer: (D) -2

The numerator is $(x-6)(x+2)$, but $x = 6$ also makes the denominator 0, so $h(6)$ is undefined. The only zero is $x = -2$; the sum is -2.

Trap: Students set the numerator equal to 0 and use the sum of its roots, 4.

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18. **AM-307** · Hard · Multiple choice · **No calculator needed**

In the equation $\dfrac{x^2}{x-a} = \dfrac{9}{x-a}$, a is a constant. For how many real values of a does the equation have exactly one real solution?

A) Infinitely many

B) Exactly two

C) Exactly one

D) Zero

Answer: (B) Exactly two

Multiplying gives $x^2 = 9$, so $x = \pm 3$, with the restriction $x \neq a$. The equation has exactly one solution when a equals one of these roots: $a = 3$ or $a = -3$. Two values.

Trap: Students treat the two roots as fixed and answer 0, or think every a works.

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19. **AM-312** · Hard · Multiple choice · **No calculator needed**

How many real solutions does the equation $\sqrt{x^2 - 10x + 25} = 5 - x$ have?

A) Exactly two

B) Zero

C) Infinitely many

D) Exactly one

Answer: (C) Infinitely many

$\sqrt{x^2 - 10x + 25} = \sqrt{(x-5)^2} = |x - 5|$. And $|x - 5| = 5 - x$ exactly when $x - 5 \leq 0$, that is, for every $x \leq 5$. Infinitely many solutions.

Trap: Students write $\sqrt{(x-5)^2} = x - 5$ and get only $x = 5$, or square both sides, reach $0 = 0$ and panic.

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20. **AM-314** · Hard · Multiple choice · **No calculator needed**

The expression $\dfrac{6x^2+7x-5}{2x+3}$ is equivalent to $Ax+B+\dfrac{C}{2x+3}$, where A, B and C are constants. What is the value of $A+B+C$?

A) -4

B) 2

C) 4

D) 0

Answer: (D) 0

$(2x+3)(3x-1)=6x^2+7x-3$, so $6x^2+7x-5=(2x+3)(3x-1)-2$. $A=3$, $B=-1$, $C=-2$; sum $=0$.

Trap: Students use synthetic division with $-\frac{3}{2}$ and forget to divide the quotient by 2, getting $A=6$, $B=-2$.

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21. **AM-316** · Hard · Multiple choice · **No calculator needed**

For all real numbers x, $f(2x-1)=4x^2+3$. Which of the following defines $f(x)$?

A) $f(x)=x^2+4$

B) $f(x)=x^2+2x+4$

C) $f(x)=4x^2+3$

D) $f(x)=4x^2-4x+4$

Answer: (B) $f(x)=x^2+2x+4$

Let $u=2x-1$, so $x=\dfrac{u+1}{2}$ and $4x^2=(u+1)^2$. Then $f(u)=(u+1)^2+3=u^2+2u+4$. Check: $x=3$ gives $f(5)=39$ and $25+10+4=39$ ✓.

Trap: Students copy the right side unchanged, or substitute $2x-1$ into it instead of inverting.

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22. **AM-320** · Hard · SPR · **No calculator needed**

In the equation $\dfrac{x+a}{x^2-a^2}=-\dfrac{3}{20x}$, a is a constant. If $x=6$ is a solution, what is the value of a?

Answer: 46

For $x\neq\pm a$, $\dfrac{x+a}{(x+a)(x-a)}=\dfrac{1}{x-a}$. So $\dfrac{1}{x-a}=-\dfrac{3}{20x}\Rightarrow 20x=-3x+3a\Rightarrow a=\dfrac{23x}{3}=46$. Check: $\dfrac{52}{36-2116}=-\dfrac{1}{40}=-\dfrac{3}{120}$ ✓.

Trap: Students cross-multiply the full quadratic and lose signs, or stop at a value of x.

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23. **AM-322** · Hard · Multiple choice · **No calculator needed**

In the equation $\dfrac{ax}{x-2}=3+\dfrac{4}{x-2}$, a is a constant. For which values of a does the equation have no solution?

A) 3 and 4 only

B) 3 only

C) 2 only

D) 2 and 3 only

Answer: (D) 2 **and** 3 **only**

Multiply by $x-2$: $ax=3(x-2)+4$, so $(a-3)x=-2$. If $a=3$: $0=-2$, no solution. Otherwise $x=\dfrac{-2}{a-3}$, which is extraneous when it equals 2: $a-3=-1$, $a=2$. So $a=2$ and $a=3$.

Trap: Students find only the coefficient case ($a=3$) and forget the extraneous-root case.

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24. AM-324 · Hard · SPR · **No calculator needed**

For how many integer values of b does the equation $x^2 + bx + 36 = 0$ have two distinct integer solutions?

Answer: 8

The roots p, q satisfy $pq = 36$ and $b = -(p+q)$. Distinct factor pairs: $(1, 36), (2, 18), (3, 12), (4, 9)$ and their negatives, giving $b = \pm 37, \pm 20, \pm 15, \pm 13$: 8 values. $(6, 6)$ is a repeated root.

Trap: Students count positive b only (4) or include $b = \pm 12$ (10).

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25. AM-327 · Hard · Multiple choice · **No calculator needed**

What is the solution set of the equation $2x^{2/3} - 3 = 15$?

A) $\{-27, 27\}$

B) $\{27\}$

C) $\{-9^{2/3}, 9^{2/3}\}$

D) $\{-6, 6\}$

Answer: (A) $\{-27, 27\}$

$x^{2/3} = 9 \Rightarrow \left(x^{1/3}\right)^2 = 9 \Rightarrow x^{1/3} = \pm 3 \Rightarrow x = \pm 27$. Check: $(-27)^{2/3} = \left((-27)^{1/3}\right)^2 = 9$.

Trap: Students keep only the positive root, or invert the exponent the wrong way.

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26. AM-329 · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = 2(x-1)(x-7)$. For what value of k does the graph of $y = f(x) + k$ have exactly one x-intercept?

A) 9

B) 14

C) 18

D) -18

Answer: (C) 18

The vertex is midway between the zeros, at $x = 4$: $f(4) = 2(3)(-3) = -18$. Shifting up by 18 makes the vertex touch the x-axis: $k = 18$.

Trap: Students answer the minimum value -18, or forget the leading coefficient (9).

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27. AM-311 · Medium · SPR · **No calculator needed**

What is the sum of all solutions to the equation $(x-3)^2 = (2x+1)^2$?

Answer: $-\dfrac{10}{3}$ **(or** -3.333**)**

$x - 3 = \pm(2x + 1)$. With $+$: $x = -4$. With $-$: $x - 3 = -2x - 1 \Rightarrow x = \frac{2}{3}$. Sum $= -4 + \frac{2}{3} = -\frac{10}{3}$.

Trap: Students take square roots without $\pm$ and find only $x = -4$.

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28. AM-331 · Medium · Multiple choice · **No calculator needed**

If $2a - 3b = 7$, what is the value of $\dfrac{4^a}{8^b}$?

A) $\dfrac{1}{128}$

B) 2^{14}

C) 7

D) 128

Answer: (D) 128

$\dfrac{4^a}{8^b} = \dfrac{2^{2a}}{2^{3b}} = 2^{2a-3b} = 2^7 = 128$.

Trap: Students answer 7, or subtract the exponents in the wrong order.

Geometry and Trigonometry (21 items)

29. GT-255 · Hard · Multiple choice · **Calculator**

Model Earth as a sphere with radius $6{,}371$ kilometers. An observer's eye is 1 kilometer above the surface. The line of sight from the eye to the horizon is tangent to Earth. Which is closest to the distance, in kilometers, from the eye to the point where the line of sight touches Earth?

A) 112.9

B) 1.0

C) $6{,}371.0$

D) 79.8

Answer: (A) 112.9

The radius to the point of tangency is perpendicular to the line of sight: $d = \sqrt{6{,}372^2 - 6{,}371^2} = \sqrt{12{,}743} \approx 112.9$.

Trap: Students compute $\sqrt{Rh} \approx 79.8$ (dropping the factor 2 in $2Rh$) or do not see the right triangle.

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30. GT-232 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, a circle has equation $(x-4)^2 + (y+2)^2 = 50$. Which point lies in the interior of the circle?

A) $(10, 2)$

B) $(9, 3)$

C) $(-1, -6)$

D) $(-3, 0)$

Answer: (C) $(-1, -6)$

Compare $(x-4)^2 + (y+2)^2$ with 50: $(-3, 0) \to 53$; $(9, 3) \to 50$ (on the circle); $(10, 2) \to 52$; $(-1, -6) \to 25 + 16 = 41 < 50$, inside.

Trap: Students check each coordinate separately ($|x-4| < \sqrt{50}$ and $|y+2| < \sqrt{50}$), choose a point that is on the circle, or flip the centre's signs.

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31. GT-234 · Hard · Multiple choice · **No calculator needed**

For $0 < t < \dfrac{\pi}{2}$, $\sin t = 0.6$. What is the value of $\cos(\pi + t) + \sin(\pi - t)$?

A) 0.2

B) -1.4

C) -0.2

D) 1.4

Answer: (C) -0.2

$\cos t = 0.8$. On the unit circle, $\pi + t$ is in Quadrant III, so $\cos(\pi + t) = -\cos t = -0.8$; $\pi - t$ is in Quadrant II, so $\sin(\pi - t) = \sin t = 0.6$. Sum $= -0.2$.

Trap: Students make both terms negative (-1.4) or both positive (1.4).

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32. GT-236 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that the measure of angle AED equals the measure of angle ABC. If $AD = 4$, $DB = 11$ and $AE = 6$, what is the length of $\overline{EC}$?

A) 10

B) 16.5

C) 22.5

D) 4

Answer: (D) 4

Triangles AED and ABC share angle A and have $\angle AED = \angle ABC$, so $A \leftrightarrow A$, $E \leftrightarrow B$, $D \leftrightarrow C$. Then $\dfrac{AE}{AB} = \dfrac{AD}{AC}$: $\dfrac{6}{15} = \dfrac{4}{AC}$, $AC = 10$, $EC = 10 - 6 = 4$. $\overline{DE}$ is NOT parallel to $\overline{BC}$.

Trap: Students assume $\overline{DE} \parallel \overline{BC}$ and use $\dfrac{AD}{DB} = \dfrac{AE}{EC}$, getting 16.5.

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33. GT-239 · Hard · SPR · **No calculator needed**

A rhombus has side length 13 and area 120. What is the sum of the lengths of its diagonals?

Answer: 34

Area $= \frac{1}{2}d_1d_2 = 120 \Rightarrow d_1d_2 = 240$. Half-diagonals and a side form a right triangle: $d_1^2 + d_2^2 = 4(13^2) = 676$. $(d_1 + d_2)^2 = 676 + 480 = 1156$, so $d_1 + d_2 = 34$ (diagonals 24 and 10).

Trap: Students forget the $\frac{1}{2}$ in the area formula or answer $2 \cdot 13 = 26$.

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34. GT-241 · Hard · SPR · **No calculator needed**

A sector of a circle has radius 10 and central angle $216°$. The sector is rolled into a right circular cone so that its two straight edges meet. The volume of the cone is $k\pi$. What is the value of k?

Answer: 96

Arc length $= 10 \cdot \dfrac{216}{360} \cdot 2\pi = 12\pi$, which is the base circumference, so $r = 6$. The sector radius is the slant height: $h = \sqrt{10^2 - 6^2} = 8$. $V = \frac{1}{3}\pi(36)(8) = 96\pi$.

Trap: Students use 10 as the cone's radius or height.

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35. GT-243 · Hard · Multiple choice · **No calculator needed**

In triangles PQR and XYZ, $PQ = XY = 9$ and $m\angle R = m\angle Z = 35°$, where $\overline{PQ}$ is opposite angle R and $\overline{XY}$ is opposite angle Z. Which additional fact is enough to prove that triangle PQR is congruent to triangle XYZ?

A) $PR = XZ$

B) $QR = YZ = 14$

C) $m\angle Q = m\angle Y$

D) $PR = QR$

Answer: (C) $m\angle Q = m\angle Y$

With $\angle Q = \angle Y$, the triangles have two angles and a non-included side equal (AAS). Choices with one more side give SSA, which can fit two different triangles (for $QR = 14$ there are two triangles, with $PR \approx 15.5$ or $PR \approx 7.4$). $PR = QR$ is about one triangle only.

Trap: Students accept side-side-angle as a congruence test.

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36. GT-246 · Hard · Multiple choice · **No calculator needed**

Triangle ABC has area 60. Point D lies on $\overline{BC}$ with $BD : DC = 2 : 3$, and point E lies on $\overline{AD}$ with $AE : ED = 3 : 1$. What is the area of triangle BED?

A) 15

B) 18

C) 10

D) 6

Answer: (D) 6

Triangles ABD and ABC share the height from A: area $ABD = \frac{2}{5} \cdot 60 = 24$. Triangles BED and BAD share the height from B: area $BED = \frac{1}{4} \cdot 24 = 6$.

Trap: Students use the wrong part of a ratio ($\frac{3}{4}$ instead of $\frac{1}{4}$, or $\frac{2}{3}$ instead of $\frac{2}{5}$).

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37. GT-249 · Hard · Multiple choice · **No calculator needed**

Sphere Q has volume 36π. Sphere P has a radius 7 times the radius of sphere Q. The spheres are externally tangent at point T. Segment $\overline{MT}$ is a diameter of sphere P and segment $\overline{TN}$ is a diameter of sphere Q, with M, T and N on one line. What is the length of $\overline{MN}$?

A) 66

B) 45

C) 24

D) 48

Answer: (D) 48

$\frac{4}{3}\pi r^3 = 36\pi \Rightarrow r^3 = 27$, $r = 3$. Sphere P has radius 21. $MN = 2(21) + 2(3) = 48$.

Trap: Students add the radii (24) or solve $r^2 = 27$.

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38. GT-252 · Hard · SPR · **No calculator needed**

Two identical right rectangular prisms each have a square base with side length s and a height of 35. Each prism has surface area K. The prisms are glued together base to base, and the new solid has surface area $\frac{15}{8}K$. What is the value of s?

Answer: 10

$K = 2s^2 + 140s$. Gluing hides one base on each prism: new area $= 2K - 2s^2$. So $2K - 2s^2 = \frac{15}{8}K \Rightarrow \frac{1}{8}K = 2s^2 \Rightarrow K = 16s^2$. Then $2s^2 + 140s = 16s^2 \Rightarrow s = 10$.

Trap: Students subtract the glued face once, or set the new area to $2K$.

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39. GT-256 · Hard · Multiple choice · **No calculator needed**

Parallel lines m and n are cut by a transversal. Angles b and c are same-side interior angles, and angle a is one of the other angles formed. If $200 < a + b + c < 236$, where all measures are in degrees, which of the following could be the measure of angle a, in degrees?

A) 56

B) 18

C) 64

D) 40

Answer: (D) 40

Same-side interior angles are supplementary: $b + c = 180$. So $200 < a + 180 < 236$, giving $20 < a < 56$. Only 40 works (56 is excluded by the strict inequality; 18 and 64 are outside).

Trap: Students include an endpoint or do not use $b + c = 180$.

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40. GT-258 · Hard · Multiple choice · **No calculator needed**

Right triangles PQR and XYZ are similar, with P, Q and R corresponding to X, Y and Z, and right angles at Q and Y. If $\tan P = \frac{8}{15}$ and $XZ = 51$, what is the length of $\overline{XY}$?

A) 27.2

B) 45

C) 24

D) 95.625

Answer: (B) 45

$\tan X = \tan P = \frac{8}{15} = \frac{YZ}{XY}$, so the sides are $8k$, $15k$ and hypotenuse $17k$. $17k = 51$, $k = 3$, $XY = 15k = 45$.

Trap: Students give the opposite side (24) or multiply the hypotenuse by the tangent (27.2) or divide by it (95.625).

---

41. GT-260 · Hard · Multiple choice · **No calculator needed**

A rectangular box has a base that measures 9 by 12 and a height of 20. Let θ be the angle between a diagonal of the box (from a bottom corner to the opposite top corner) and the base. What is the value of $\tan\theta$?

A) $\dfrac{20}{9}$

B) $\dfrac{5}{3}$

C) $\dfrac{3}{4}$

D) $\dfrac{4}{3}$

Answer: (D) $\dfrac{4}{3}$

The diagonal's projection on the base is the base diagonal, $\sqrt{9^2+12^2}=15$. $\tan\theta=\dfrac{20}{15}=\dfrac{4}{3}$.

Trap: Students use one base edge (9 or 12) instead of the base diagonal, or flip the ratio.

---

42. GT-262 · Hard · Multiple choice · **No calculator needed**

Right triangle ABC has its right angle at C, $AC=9$ and $BC=12$. Each leg is lengthened by 3 and the angle at C stays a right angle. What is $\tan A$ in the new triangle?

A) $\dfrac{4}{5}$

B) $\dfrac{4}{3}$

C) $\dfrac{5}{4}$

D) $\dfrac{5}{3}$

Answer: (C) $\dfrac{5}{4}$

New legs: $AC=12$, $BC=15$. $\tan A=\dfrac{BC}{AC}=\dfrac{15}{12}=\dfrac{5}{4}$. Adding the same amount to each leg does not keep the triangle similar, so the ratio changes from $\dfrac{4}{3}$.

Trap: Students assume the trig ratio is unchanged ($\frac{4}{3}$).

---

43. GT-264 · Hard · Multiple choice · **No calculator needed**

A drone flies 12 kilometers in a direction $60°$ north of east, then flies 12 kilometers due west. How far, in kilometers, is the drone from its starting point?

A) 12

B) 24

C) $12\sqrt{2}$

D) $6\sqrt{3}$

Answer: (A) 12

First leg ends at $(12\cos 60°, 12\sin 60°)=(6, 6\sqrt{3})$. After moving 12 west: $(-6, 6\sqrt{3})$. Distance $=\sqrt{36+108}=12$.

Trap: Students assume the legs are perpendicular ($12\sqrt{2}$) or add them (24).

---

44. GT-266 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the points $A(0,0)$, $B(6,3)$ and $C(4,7)$ are the vertices of a triangle. What is the value of $\tan(\angle CAB)$?

A) $\frac{1}{2}$

B) $\frac{2}{3}$

C) $\frac{3}{2}$

D) $\frac{7}{4}$

Answer: (B) $\frac{2}{3}$

Slopes: $AB = \frac{1}{2}$, $BC = \frac{7-3}{4-6} = -2$; their product is -1, so the right angle is at B. $AB = \sqrt{45}$, $BC = \sqrt{20}$, $\tan A = \frac{BC}{AB} = \sqrt{\frac{20}{45}} = \frac{2}{3}$.

Trap: Students give the slope of $\overline{AC}$ ($\frac{7}{4}$) or of $\overline{AB}$ ($\frac{1}{2}$) instead of a ratio of sides.

---

45. GT-269 · Hard · Multiple choice · **No calculator needed**

In a right triangle, the measure of acute angle A is between $45°$ and $90°$. Which of the following is true?

A) $\cos A < \sin A < \tan A$

B) $\tan A < \sin A < \cos A$

C) $\sin A < \cos A < \tan A$

D) $\cos A < \tan A < \sin A$

Answer: (A) $\cos A < \sin A < \tan A$

Above $45°$ the opposite leg is longer than the adjacent leg, so $\cos A < \sin A$, and $\tan A = \frac{\sin A}{\cos A} > 1 > \sin A$. Check $60°$: $0.5 < 0.866 < 1.732$.

Trap: Students use the ordering that holds below $45°$ ($\sin A < \cos A$).

---

46. GT-271 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, line ℓ has slope $\frac{3}{4}$ and line k is perpendicular to line ℓ. Line k makes an angle θ with the positive x-axis, measured counterclockwise, where $0° < \theta < 180°$. What is the value of $\cos\theta$?

A) $-\frac{4}{5}$

B) $\frac{3}{5}$

C) $-\frac{3}{5}$

D) $\frac{4}{5}$

Answer: (C) $-\frac{3}{5}$

Slope of k: $-\frac{4}{3}$, so θ is obtuse with direction $(-3,4)$ (length 5). $\cos\theta = -\frac{3}{5}$.

Trap: Students drop the negative sign or use the sine ($\frac{4}{5}$).

---

47. GT-273 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $52°$. The bisectors of angles B and C intersect at point I. What is the measure, in degrees, of angle BIC?

A) 116

B) 128

C) 64

D) 104

Answer: (A) 116

$B + C = 128°$, so $\frac{B}{2} + \frac{C}{2} = 64°$ and $\angle BIC = 180° - 64° = 116°$.

Trap: Students compute $180 - 52 = 128$, or $2 \cdot 52 = 104$, or stop at 64.

---

48. GT-238 · Medium · Multiple choice · **No calculator needed**

A triangle has two sides of lengths 10 and 12, and the angle between them measures $150°$. What is the area of the triangle?

A) $30\sqrt{3}$

B) 30

C) 15

D) 60

Answer: (B) 30

The height to the side of length 10 is $12 \sin 150° = 12 \cdot \frac{1}{2} = 6$ (it falls outside the triangle). Area $= \frac{1}{2} \cdot 10 \cdot 6 = 30$.

Trap: Students use $\sin 60°$ or treat the triangle as right ($\frac{1}{2} \cdot 10 \cdot 12 = 60$).

---

49. GT-267 · Medium · Multiple choice · **No calculator needed**

A parallelogram has sides of lengths 12 and 20. The cosine of one of its acute angles is 0.8. What is the area of the parallelogram?

A) 72

B) 144

C) 192

D) 240

Answer: (B) 144

$\sin\theta = \sqrt{1 - 0.8^2} = 0.6$. Height to the side of length 20 is $12 \cdot 0.6 = 7.2$. Area $= 20 \cdot 7.2 = 144$.

Trap: Students use the cosine as the height ratio (192) or ignore the angle (240).

Problem-Solving and Data Analysis (27 items)

50. PS-164 · Hard · Multiple choice · **Calculator**

Electric vehicles made up 4% of a country's new-car sales in one year and 6% the next year. Total new-car sales increased by 10% from the first year to the second. By what percent did the number of electric vehicles sold increase?

A) 2%

B) 60%

C) 65%

D) 50%

Answer: (C) 65%

Let first-year sales be 100: EVs $= 4$. Second year: 110 cars, EVs $= 0.06(110) = 6.6$. Increase $= \dfrac{2.6}{4} = 65\%$.

Trap: Students answer 2 (percentage points) or 50 (ignoring the change in total sales).

---

51. PS-167 · Hard · SPR · **Calculator**

Bottle B holds 20% more juice than bottle A and costs 15% more. To the nearest tenth of a percent, by what percent is the price per liter of bottle B less than the price per liter of bottle A?

Answer: 4.2

Price per liter of B relative to A: $\dfrac{1.15}{1.20} \approx 0.9583$. It is about 4.2% less.

Trap: Students subtract the percents and answer 5.

---

52. PS-179 · Hard · Multiple choice · **Calculator**

A restaurant adds a 15% tax and a 10% service charge to a bill. Both are calculated on the original subtotal. The total paid is 250 dollars. What was the subtotal, in dollars?

A) 200.00

B) 217.39

C) 197.63

D) 187.50

Answer: (A) 200.00

Both percents use the same base, so they add: total $= 1.25 \cdot$subtotal. Subtotal $= \dfrac{250}{1.25} = 200$.

Trap: Students chain the percents ($\frac{250}{1.15 \cdot 1.10} \approx 197.63$) or ignore one charge.

---

53. PS-182 · Hard · Multiple choice · **Calculator**

Two positive numbers a and b are in the ratio $a : b = 9 : 5$. The number b is what percent less than a, to the nearest tenth of a percent?

A) 44.4%

B) 80.0%

C) 55.6%

D) 28.6%

Answer: (A) 44.4%

$\dfrac{a-b}{a} = \dfrac{4}{9} \approx 44.4\%$.

Trap: Students use b as the base (80%, which is how much greater a is) or use the total parts.

---

54. PS-157 · Hard · Multiple choice · **No calculator needed**

Data sets A and B each contain 20 values, all from the list $10, 20, 30, 40, 50$. The frequencies in A are $2, 3, 10, 3, 2$ and the frequencies in B are $5, 3, 4, 3, 5$, in that order. Which statement is true?

A) The standard deviations are equal because the means and ranges are equal.

B) The standard deviation of B is greater.

C) The standard deviation of A is greater.

D) The standard deviations cannot be compared without calculating them.

Answer: (B) The standard deviation of B is greater.

Both sets have mean 30 and range 40. B has more values at the extremes (10 and 50) and fewer at the mean, so its values are farther from the mean on average: SD of B (≈ 15.2) is greater than SD of A (≈ 10.5).

Trap: Students think equal means and ranges force equal standard deviations.

---

55. PS-160 · Hard · SPR · **No calculator needed**

The data set $2, 6, 9, 15, x$ has a mean equal to its median, where x is a real number. What is the sum of all possible values of x?

Answer: 19

Mean $= \dfrac{32+x}{5}$. If $x \le 6$, median 6: $x = -2$ ✓. If $6 \le x \le 9$, median x: $32 + x = 5x$, $x = 8$ ✓. If $x \ge 9$, median 9: $x = 13$ ✓. Sum $= -2 + 8 + 13 = 19$.

Trap: Students assume x is the median, or reject the negative value.

---

56. PS-162 · Hard · Multiple choice · **No calculator needed**

The commute times, in whole minutes, of 20 students are grouped: 10 students took 0 to 9 minutes, 7 took 10 to 19 minutes, and 3 took 20 to 29 minutes. Which of the following could NOT be the median commute time?

A) 10

B) 7.5

C) 14

D) 4

Answer: (D) 4

The median is the mean of the 10th and 11th values. The 10th is in 0–9 and the 11th is in 10–19, so the median is between $\frac{0+10}{2} = 5$ and $\frac{9+19}{2} = 14$. 4 is impossible; 7.5, 10 and 14 are possible.

Trap: Students think the median must lie inside one bin, or that it cannot be bounded at all.

---

57. PS-165 · Hard · SPR · **No calculator needed**

The number a is 150% more than the number b, and b is 60% less than the positive number c. The number a is what percent of c?

Answer: 100

$a = 2.5b$ and $b = 0.4c$, so $a = c$: 100%.

Trap: Students read "150% more than" as $1.5b$, giving 60%.

---

58. PS-169 · Hard · SPR · **No calculator needed**

In grade 10, 12 students are in a club and 28 are not. In grade 11, k students are in the club and 42 are not. Being in the club is independent of grade. What is the value of k?

Answer: 18

Independence means the proportion in the club is the same in each grade: $\dfrac{12}{40} = \dfrac{k}{k+42}$. $0.3k + 12.6 = k \Rightarrow k = 18$.

Trap: Students set $k = 12$ (equal counts) or compare with the wrong total.

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59. PS-171 · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.5x + 40$. For the data point with $x = 24$, the predicted value of y is 20% less than the actual value. What is the residual for this data point?

Answer: 25

Predicted $= 2.5(24) + 40 = 100$. $100 = 0.8 \cdot$actual, so actual $= 125$. Residual $=$ actual $-$ predicted $= 25$.

Trap: Students take 20% of the predicted value (20) or get the sign of the residual wrong.

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60. PS-174 · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.4x + 15$. A new data set is formed by adding 5 to every x-value and keeping each y-value. What is the y-intercept of the line of best fit for the new data set?

Answer: 3

Shifting every x right by 5 shifts the line right by 5: $\hat{y} = 2.4(x-5) + 15 = 2.4x + 3$. Same slope, new intercept 3.

Trap: Students add 5 to the intercept (20) or change the slope.

---

61. PS-176 · Hard · Multiple choice · **No calculator needed**

If 18% of a is 24, what is $a\%$ of 18?

A) 4.32

B) 24

C) $2{,}400$

D) $133\frac{1}{3}$

Answer: (B) 24

$a\%$ of $18 = \dfrac{a \cdot 18}{100} = 18\%$ of $a = 24$. No need to find a.

Trap: Students solve for a ($133\frac{1}{3}$) and stop, or multiply $0.18 \cdot 24$.

---

62. PS-183 · Hard · Multiple choice · **No calculator needed**

The number y is $p\%$ less than the positive number x, where $0 < p < 100$. Which expression is equal to x?

A) $\dfrac{(100-p)y}{100}$

B) $\dfrac{100y}{100+p}$

C) $y\left(1+\dfrac{p}{100}\right)$

D) $\dfrac{100y}{100-p}$

Answer: (D) $\dfrac{100y}{100-p}$

$y = x\left(1-\dfrac{p}{100}\right) = x \cdot \dfrac{100-p}{100}$, so $x = \dfrac{100y}{100-p}$. Check $p = 20$, $y = 80$: $x = 100$.

Trap: Students add $p\%$ back to y (using the wrong base).

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63. PS-185 · Hard · Multiple choice · **No calculator needed**

In a survey, exactly 37.5% of the people answered "yes," exactly 20% answered "no," and the rest were undecided. What is the least possible number of people surveyed?

A) 5

B) 40

C) 8

D) $1{,}000$

Answer: (B) 40

$37.5\% = \dfrac{3}{8}$ and $20\% = \dfrac{1}{5}$, so the number surveyed must be a multiple of 8 and of 5: least is 40 (15 yes, 8 no, 17 undecided).

Trap: Students satisfy only one percent (8 or 5).

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64. PS-188 · Hard · Multiple choice · **No calculator needed**

A pipe can fill an empty tank in 6 hours, and a drain can empty the full tank in 9 hours. Starting with the tank empty, the pipe and the drain are both open for 4 hours, and then the drain is closed. How many more hours does it take the pipe to fill the tank?

A) 2

B) $\dfrac{26}{3}$

C) $\dfrac{14}{3}$

D) 14

Answer: (C) $\frac{14}{3}$

Net rate $\frac{1}{6}-\frac{1}{9}=\frac{1}{18}$ tank per hour; after 4 hours: $\frac{2}{9}$ full. Remaining $\frac{7}{9}$ at $\frac{1}{6}$ per hour: $\frac{7/9}{1/6}=\frac{14}{3}$ hours (4 h 40 min).

Trap: Students give the total time ($8\frac{2}{3}$ h) or leave the drain open.

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65. PS-190 · Hard · Multiple choice · **No calculator needed**

A driver travels from one city to another at an average speed of 60 kilometers per hour. At what average speed, in kilometers per hour, must the driver return along the same road so that the average speed for the whole round trip is 80 kilometers per hour?

A) 120

B) 106.7

C) 160

D) 100

Answer: (A) 120

Let the distance be d. $\frac{2d}{d/60+d/v}=80 \Rightarrow \frac{2}{80}=\frac{1}{60}+\frac{1}{v} \Rightarrow \frac{1}{v}=\frac{1}{120}$, $v=120$.

Trap: Students average the speeds (100).

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66. PS-193 · Hard · Multiple choice · **No calculator needed**

The data set $3,5,6,8,9,11,12,46$ has one outlier, 46, which is removed. Which statement about the effect of removing 46 is true?

A) Only the mean changes.

B) Both the mean and the median decrease, and the mean decreases more.

C) The mean decreases and the median increases.

D) The mean decreases and the median stays the same.

Answer: (B) Both the mean and the median decrease, and the mean decreases more.

Mean: $12.5 \to \frac{54}{7} \approx 7.71$. Median: $\frac{8+9}{2}=8.5 \to 8$ (the count changes from 8 to 7). Both decrease, and the mean decreases more.

Trap: Students think the median cannot change when an outlier is removed.

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67. PS-195 · Hard · Multiple choice · **No calculator needed**

Section P has 60 students and section Q has 24 students. Test-score summaries: Section P: minimum 55, first quartile 68, median 76, third quartile 84, maximum 97. Section Q: minimum 62, first quartile 77, median 83, third quartile 88, maximum 95. Which statement must be true?

A) At least 75% of the students in Q scored above the median of P.

B) The IQR of Q is greater than the IQR of P.

C) More students in Q than in P scored above 84.

D) The range of Q is greater than the range of P.

Answer: (A) At least 75% of the students in Q scored above the median of P.

In Q, at least 75% of the scores are at or above its first quartile, 77, which is above P's median, 76. the statement about more students above 84 cannot be decided (different section sizes; only percents are known); the range of Q (33) is less than that of P (42); the IQR of Q (11) is less than that of P (16).

Trap: Students compare counts when only percents are known, or compare ranges by eye.

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68. PS-197 · Hard · SPR · **No calculator needed**

In a data set, the values 1, 2, 3, 4 and 5 occur with relative frequencies 10%, 15%, 20%, 30% and 25%, respectively. What is the median of the data set minus the mean of the data set?

Answer: 0.55 **(or** $\frac{11}{20}$**)**

Cumulative: $10\%, 25\%, 45\%, 75\%$, so the 50% point is at the value 4: median $= 4$. Mean $= 0.1+0.3+0.6+1.2+1.25 = 3.45$. Difference $= 0.55$.

Trap: Students think the total count is needed, or take the median of the percents.

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69. PS-199 · Hard · Multiple choice · **No calculator needed**

A data set of 30 values has a mean of 50 and a standard deviation of 10. The values 38 and 62 are added to the data set. Which statement is true?

A) The mean is unchanged and the standard deviation increases.

B) The mean is unchanged and the standard deviation decreases.

C) The mean and the standard deviation are both unchanged.

D) The mean and the standard deviation both increase.

Answer: (A) The mean is unchanged and the standard deviation increases.

The pair is symmetric about 50, so the mean stays 50. Each new value is 12 from the mean, farther than the typical distance 10, so the standard deviation increases (to about 10.1).

Trap: Students assume values inside the range lower the SD, or that a symmetric pair leaves it unchanged.

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70. PS-202 · Hard · Multiple choice · **No calculator needed**

A company randomly selected 200 of its employees and gave them new office chairs. Their average productivity rose 8% during the next 3 months. Which conclusion is best supported?

A) The new chairs caused the rise for all employees.

B) The rise cannot be attributed to the chairs, because there was no comparison group.

C) The new chairs had no effect on productivity.

D) The new chairs caused the rise for the 200 employees only.

Answer: (B) The rise cannot be attributed to the chairs, because there was no comparison group.

There was no comparison group without new chairs, so the rise could come from other changes over the same months. Random selection allows generalization, not a causal claim.

Trap: Students think a random sample alone supports cause and effect.

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71. PS-204 · Hard · Multiple choice · **No calculator needed**

A survey was mailed to $1{,}200$ randomly selected households, and 180 surveys were returned. Of these, 72% supported a new fee. What is the main reason the 72% may not represent all households?

A) There was no random assignment to treatments.

B) The households were selected at random.

C) The sample of $1{,}200$ is too small.

D) Many selected households did not respond, and those who responded may differ from those who did not.

Answer: (D) Many selected households did not respond, and those who responded may differ from those who did not.

Only 15% responded. People who choose to return a survey may differ from those who do not (non-response bias), even though the original selection was random.

Trap: Students think random selection guarantees a representative result.

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72. PS-210 · Hard · Multiple choice · **No calculator needed**

A model for a data set is $\hat{y} = 2.5x + 10$. The data points are $(2, 17)$, $(4, 19.5)$, $(6, 24)$, $(8, 31)$ and $(10, 34.5)$. For how many of the data points does the model underestimate the actual value of y?

A) 5

B) 2

C) 3

D) 1

Answer: (B) 2

Residuals (actual − predicted): $17 - 15 = 2$, $19.5 - 20 = -0.5$, $24 - 25 = -1$, $31 - 30 = 1$, $34.5 - 35 = -0.5$. The model underestimates when the residual is positive: 2 points.

Trap: Students reverse "underestimate" and count the negative residuals (3).

---

73. PS-187 · Medium · Multiple choice · **Calculator**

A runner's pace is 6 minutes 15 seconds per kilometer. What is the runner's speed, in kilometers per hour?

A) 9.6

B) 6.25

C) 9.76

D) 0.16

Answer: (A) 9.6

6 min 15 s $= 6.25$ min. Speed $= \dfrac{60}{6.25} = 9.6$ km/h.

Trap: Students read 6 min 15 s as 6.15 min (≈ 9.76).

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74. PS-173 · Medium · SPR · **No calculator needed**

The ratio of grade 9 students to grade 10 students in a club is $3 : 5$, and the ratio of grade 10 students to grade 11 students is $4 : 7$. The club has 201 students in grades 9, 10 and 11. How many more grade 11 students than grade 9 students are in the club?

Answer: 69

Make the grade 10 terms equal: $12 : 20$ and $20 : 35$, so $12 : 20 : 35$ (sum 67). $201 \div 67 = 3$: 36, 60 and 105. Difference $= 105 - 36 = 69$.

Trap: Students write $3 : 5 : 7$ (sum 15) and divide 201 by 15.

---

75. PS-205 · Medium · Multiple choice · **No calculator needed**

An observational study found that children from homes with more books tend to have higher reading scores. Which of the following is the most plausible confounding variable?

A) Parents' education level

B) The scale used for the reading test

C) The number of pages per book

D) The number of children in the study

Answer: (A) Parents' education level

A confounder is related to both variables: parents' education level can affect both the number of books at home and the children's reading scores.

Trap: Students choose a variable linked to only one of the two variables, or a feature of the study design.

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76. **PS-207** · Medium · Multiple choice · **No calculator needed**

A teacher randomly selected 50 students and asked each one, face to face, “Have you ever copied homework?” Of the students, 6% answered yes. Which is the most likely problem with this result?

A) The sample was not selected at random.

B) There was no control group.

C) Students may give untrue answers because the question is sensitive and asked in person.

D) A sample of 50 students is too large.

Answer: (C) Students may give untrue answers because the question is sensitive and asked in person.

The question is sensitive and asked in person by a teacher, so students may under-report (response bias), even though the sample was random.

Trap: Students check only how the sample was chosen and miss the measurement problem.
